



Optical and gamma-ray shielding features of Nd³⁺ doped lithium-zinc-borophosphate glasses

Mohammed A. Algradee^{a,c,*}, Emran Eisa Saleh^b, Tharwat M. EL Sherbini^c,
R. El-Mallawany^d

^a Physics Department, Faculty of Science, Ibb University, Ibb 70270, Yemen

^b Physics Department, Faculty of Science, Cairo University, Aden, Yemen

^c Laboratory of Lasers and New Materials, Faculty of Science, Cairo University, Giza 12613, Egypt

^d Physics Dept., Faculty of Science, Menoufia University, Shebin El Koom 32511, Egypt

ARTICLE INFO

Keywords:

LZBP:Nd glasses
Structure properties
Optical properties
γ-ray shielding

ABSTRACT

Lithium-zinc-borophosphate glasses with the composition of (100-x)[20Li₂O-10ZnO-20B₂O₃-50P₂O₅]-xNd₂O₃, x = 0.0 (LZBP:0.0Nd) to 4.0 (LZBP:4.0Nd) wt%, were synthesized. DSC curves were used to determine T_g of the samples. The density of the samples increased from 2.528 to 2.588 g.cm⁻³. The physical properties such as ion concentration and field strength were estimated using standard formulae. Both of UV-visible and FTIR spectra were recorded in the range of 190–1000 nm and 400–4000 cm⁻¹, respectively. The indirect optical band gap (E_{opt}^{ASF,I}) measured by ASF method were in the range of 3.36–3.66 eV. FTIR spectra revealed the presence of PO₄, BO₃ and PO₄ units within glass matrix. The radiation shielding parameters such as the mass attenuation coefficients (MAC) and the half value layer (HVL) for the glasses were estimated via Phy-X/PSD program within the photon energy (0.015–15 MeV). The glass LZBP:4.0Nd has the lowest HVL and hence the high shielding capacity of gamma rays.

1. Introduction

Recently, several scientific groups began creating new materials based on rare earth (R.E) ions to explore their uses in various optical applications. At the forefront of these materials comes the oxides glasses which have great importance due to the ease and low costs of preparation, the availability of the raw materials involved in their composition and their wide uses. Basic Glass former include several oxides such as Si₂O, P₂O₅, B₂O₃, TeO₂, etc., and by adding some oxides to change their properties are being prepared [1].

The combination of two basic glass formers (B₂O₃ and P₂O₅) leads to form borophosphate glass, which has many important properties such as high chemical durability, narrow emission, low refractive indices, high thermal stability and low dispersion due to forming units within it called (PO₄, BO₄ and BO₃) groups [2,3].

Among the R.E. ions the trivalent Nd³⁺ ions have been extensively investigated due to their good laser efficiency and their 1.3 μm luminescence that can be used for optical amplification in the NIR region. Recently, many researchers are making more efforts to employ the glass systems as a substitute of previously radiation shielding substances such as polymers and concrete due to the glass system have high corrosion resistance. In addition, the glass system belongs to environment friendly materials and numerous

* Corresponding author at: Physics Department, Faculty of Science, Ibb University, Ibb 70270, Yemen.

E-mail addresses: algradi772001@yahoo.com, algradi772001@gmail.com (M.A. Algradee).

researchers stated that it is more suitable for radiation shielding [4].

Although there are various spectroscopic studies published on the R.E doped glasses, there is a lack of correlation between the structural, physical, optical and gamma-ray shielding features of these glasses in the literature. Therefore, the aim of the present study is to prepare the Nd³⁺-doped Lithium-zinc-borophosphate glass and to investigate the effect of Nd₂O₃ content on the structural, physical and gamma ray shielding. In addition, to determine the direct/indirect optical band gap obtained by Tauc and ASF methods, the molar refraction, the refractive index, the molar polarizability and the metallization criterion of the samples.

2. Experimental methods

Lithium-zinc-borophosphate glasses with the composition of (100-x)[20Li₂O-10ZnO-20B₂O₃-50 P₂O₅]-xNd₂O₃ with x = 0.0 (LZBP:0.0Nd) to 4.0 (LZBP:4.0Nd) wt% were synthesized via ordinary melt-quenching technique. High purity from anhydrous powders ZnO, P₂O₅, Li₂CO₃, H₂B₂O₃ and Nd₂O₃ were used. Appropriate amounts of chemicals were grinded and mixed in an agate mortar and the batches were preheated in a porcelain crucible in air at 350 °C for 1.0 h to verify the decarbonisation of lithium carbonate. The batches were melted at 1150 °C for 1.0 h and quenched on stainless steel plates and annealed in an electrical furnace at 300 °C for 1.0 h to remove the mechanical stress in the glasses. The Homogeneity of the melt was ensured by stirring regularly with an aluminum rod. The synthesized samples are marked by LZBP:0.0Nd, LZBP:0.25Nd, LZBP:0.75Nd, LZBP:2.0Nd and LZBP:4.0Nd.

The Archimede's principle was employed to measure the glass density, ρ_g , and the molar volume, V_m was computed by Eq. (1):

$$V_m = \frac{\text{Molecular weight of the glass sample}}{\rho_g} \quad (1)$$

The absorption (A) spectra were measured using a spectrophotometer Jasco V-570 Corp., in the range of wavelengths, 190–1000 nm.

Differential scanning calorimetry (DSC) measurements were carried out at a heating rate of 10 °C/min from room temperature up to 450 °C for the determination of the glass transition temperature, (T_g), using Shimadzu DSC–50 (TA Instruments). FTIR spectra of samples were measured over the range of 4000–400 cm^{−1} by JASCO FT/IR-460 plus (Japan) spectrometer with KBr disc technique.

The Nd³⁺ ion concentration, N_{Nd} , Polaron radius, r_p , Inter-ionic distance, r_i , and Field strength, F , are calculated from the next expressions (2), (3), (4) and (5) respectively, and registered in Table 1 as reported in [5,6]:

$$N_{Nd} = (\text{mol\% of Nd}) N_A \rho_g / M_w \quad (2)$$

$$r_p = \frac{1}{2} \left(\frac{\pi}{6 N_{Nd}} \right)^{1/3} \quad (3)$$

$$r_i = \left(\frac{1}{N_{Nd}} \right)^{1/3} \quad (4)$$

$$F = \frac{V_{ion}}{r_p^2} \quad (5)$$

where V_{ion} is the valence of ion.

Table 1

Density, ρ_g , Molar volume, V_m , Nd³⁺ ion concentration, N_{Nd} , Polaron radius, r_p , Inter ionic distance, r_i , Field strength, F , Packing density, V_p , Oxygen packing density, OPD , the average phosphorus-phosphorus separation, D_{P-P} , number of bonds per unit volume, n_b and glass transition temperature, T_g of glass system.

Physical parameters	Composition				
	LZBP:0 Nd	LZBP:0.25Nd	LZBP:0.75Nd	LZBP:2 Nd	LZBP:4 Nd
ρ_g (g/cm ³) ± 0.001	2.528	2.536	2.553	2.574	2.588
V_m (cm ³ /mol) ± 0.01	39.16	39.11	38.99	39.02	39.33
N_{Nd} (x 10 ²² ions/cm ³) ± 0.0001	–	0.1143	0.3435	0.9120	1.7989
r_p (Å) ± 0.001	–	3.854	2.671	1.929	1.538
r_i (Å) ± 0.001	–	9.564	6.628	4.786	3.817
F (x 10 ¹⁶) (cm ^{−2}) ± 0.001	–	0.202	0.421	0.806	1.26818
V_p ± 0.0001	0.6116	0.6124	0.6143	0.6141	0.6094
OPD (g-atom/liter) ± 0.001	86.822	86.921	87.174	87.081	86.329
D_{P-P} (x 10 ^{−2}) (Å) ± 0.0001	402.13	401.87	401.26	400.85	401.14
n_b (x10 ²⁷) (m ^{−3}) ± 0.01	64.59	64.69	64.93	64.99	64.63
T_g (°C)	378.11	382.43	386.28	403.21	420.15

For the purpose of evaluating the prepared glasses to the gamma-ray shield, the mass attenuation coefficients (MAC), the half value layer (HVL), the effective atomic number (Z_{eff}) and the electron density (N_{eff}) have been determined via Phy-X/PSD software program and using the following equations [7]:

$$\frac{\mu}{\rho} = \sum_i W_i \left(\frac{\mu}{\rho} \right)_i \quad (6)$$

$$HVL = \frac{0.693}{\mu} \quad (7)$$

$$Z_{\text{eff}} = \frac{\sum_i f_i A_i \left(\frac{\mu}{\rho} \right)_i}{\sum_j \frac{A_j}{Z_j} \left(\frac{\mu}{\rho} \right)_j} \quad (8)$$

$$N_{\text{eff}} = \frac{N_A}{N} Z_{\text{eff}} \sum n_i \quad (9)$$

where $\frac{\mu}{\rho}$ is the mass attenuation coefficient, μ is the linear attenuation coefficient and (W_i , f_i , A_i , Z_j and N_A) are the weight fraction, molar fraction, atomic weight, atomic number and Avogadro's number and $N(=\sum_i n_i A_i)$ is the atomic mass of materials.

3. Results and discussion

3.1. Glass transition temperature, T_g

DSC curves of (LZBP:0.00Nd-LZBP:4.00Nd) glasses are displayed in Fig. 1 and the glasses transition temperatures, T_g are listed in Table 1 and are shown in Fig. 2. As recorded in Table 1, T_g increased from 378.11 to 420.15 °C. The increase of T_g with increasing the amount of Nd_2O_3 may be attributed to the following reasons:

- The bond energy of Nd-O (171.29 kcal/mol) is greater than that of Li-O, Zn-O, B-O and P-O. Their bond energies are 36, 72, 119 and 88–111 kcal/mol, respectively.
- The increase in glass density and number of bonds per unit volume, n_b within glass matrix (Table 1).
- Nd^{3+} (0.983–1.35 Å) having higher ionic radius than that of other cations within glass matrix, Li^+ (0.59–0.92 Å), Zn^{2+} (0.6–0.9 Å), B^{3+} (0.01–0.27 Å) and P^{5+} (0.17–0.44 Å).

3.2. Density and molar volume

The insertion of Nd_2O_3 oxide leads to the modifications within samples networks through occupying the interstitial spaces with Nd_2O_3 which has a higher density and molecular weight ($\rho = 7.24 \text{ g.cm}^{-3}$, M.W = 336.48 g.mol^{-1}) than Li_2O ($\rho = 2.01 \text{ g.cm}^{-3}$, M.W

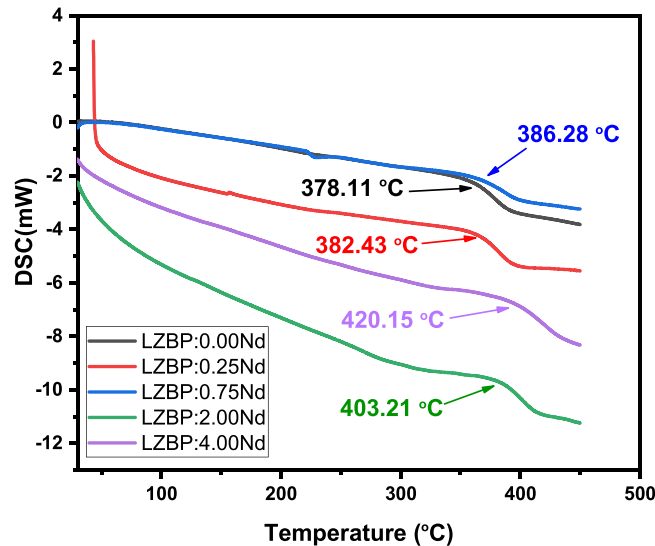


Fig. 1. DSC thermographs of Nd^{3+} -doped LZBP glasses.

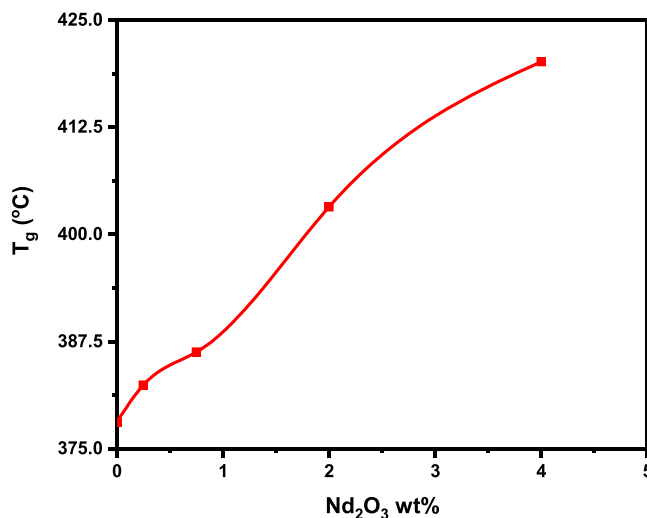


Fig. 2. Variation of glass transition temperature with Nd₂O₃ content.

= 29.88 g.mol⁻¹), ZnO ($\rho = 5.61$ g.cm⁻³, M.W = 81.38 g.mol⁻¹), B₂O₃ ($\rho = 2.46$ g.cm⁻³, M.W = 69.62 g.mol⁻¹) and P₂O₅ ($\rho = 2.39$ g.cm⁻³, M.W = 141.94 g.mol⁻¹). So that, the glass density has increased from 2.528 g.cm⁻³ to 2.588 g.cm⁻³ with increasing Nd₂O₃ content as shown in Fig. 3. Accordingly, the insertion of Nd₂O₃ into LZBP glass leads to the formation of a large number of oxygen ions inside the glass network that enhanced the transformation processes of (BO₃) units to (BO₄) units, it will be mentioned in section of FTIR spectra, where the four-coordinate boron is much denser than the three-coordinate boron. Additionally, the insertion of Nd₂O₃ leads firstly to reduce the molar volume from 39.16 to 38.99 cm³.mol⁻¹, with increasing Nd₂O₃ content from 0.0 to 0.75 wt%. Secondly, beyond 0.75 wt% of Nd₂O₃, the V_m was increased (Fig. 3) because of the large difference between ionic radius of Nd³⁺ compared to other cations within the glass system as mentioned above. Consequently, both glass compactness and rigidity are diminished because of the making of more non bridging oxygen (NBO). A significant reduction in the inter-ionic separation, D_{p-p} and Polaron radius, r_p with the rise of Nd₂O₃ content is assigned to the crowded of Nd³⁺ within glass system as listed in Table 1. Subsequently, the emergence of strong link between the Nd³⁺ and both phosphorus and boron ions leads to sturdy field strength. Molar volume displays exactly a reverse behavior to that of both packing density, V_p and oxygen packing density, OPD as listed in Table 1. Also, the addition of the Nd³⁺ ions into the LZBP glass matrix leads to the decrease of r_p and r_i between the atoms and results to increase the F value as shown in Fig. 4. Also, the Nd³⁺ ions are situated between glass layers and the Nd-O distance decrease and as a sequence, the Nd-O bond strength increase and hence create stronger F around Nd³⁺ ions. A similar trends were observed in other glass systems [5,6].

3.3. The optical band gap and Urbach energy

The UV spectra of the prepared specimens are registered in the wavelength range from 190 to 1000 nm (Fig. 5). The relation

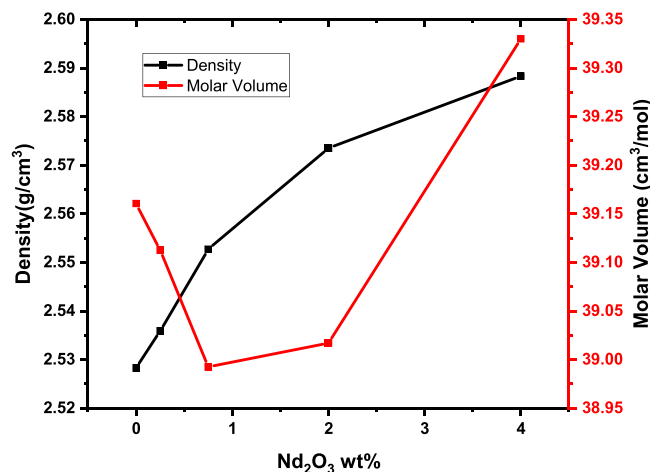


Fig. 3. Density and molar volume of glass samples against differing values of Nd₂O₃ content.

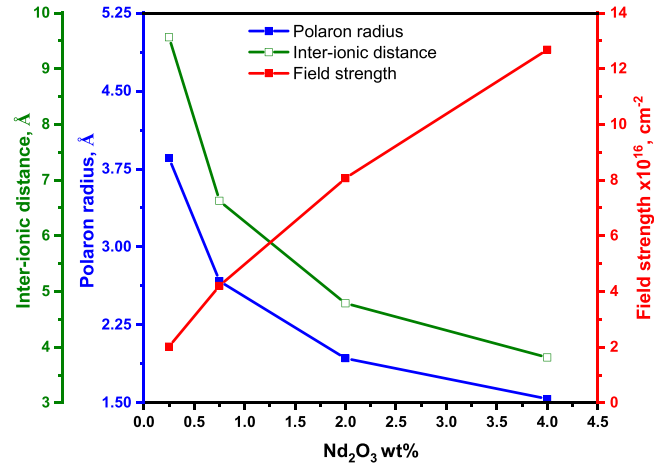


Fig. 4. r_i , r_p , and F as function of Nd_2O_3 content.

between optical absorption coefficient, $\alpha(\nu)$ and incident photon energy, $h\nu$ was proposed by Davis and Mott [8]:

$$\alpha(\nu)h\nu = [B(h\nu - E_g)]^r \quad (10)$$

where the values of the exponent r can be 2 or 1/2 for an allowed indirect transition (I) and an allowed direct transition (D) respectively, B is constant and E_g is the optical band gap. The indirect and direct E_g were obtained from Eq.(10) via the extrapolation of the linear region at $(ah\nu)^{1/2} = 0$ and $(ah\nu)^2 = 0$ respectively as shown in [Fig. 6(a,b) and 7(a,b)] (Tauc's method). The Eq.(10) can be transformed in terms of the wavelength, λ as:

$$\alpha(\lambda) = B(hc)^{r-1} \lambda \left(\lambda^{-1} - \lambda_g^{-1} \right)^r \quad (11)$$

where h is Plank's constant, c is the light velocity and λ_g is the λ related to the E_g . According to the Beer-Lambert law, the $\alpha(\nu)$ is given by:

$$\alpha(\nu) = \frac{2.303A}{d} \quad (12)$$

using Eq.(11) and Eq.(12), the $A(\lambda)$ can be obtained by [9]:

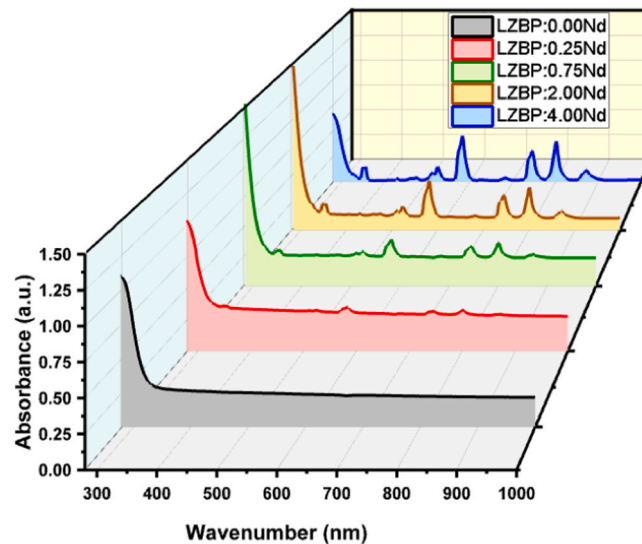


Fig. 5. Variations of optical absorption with wavelength of the samples.

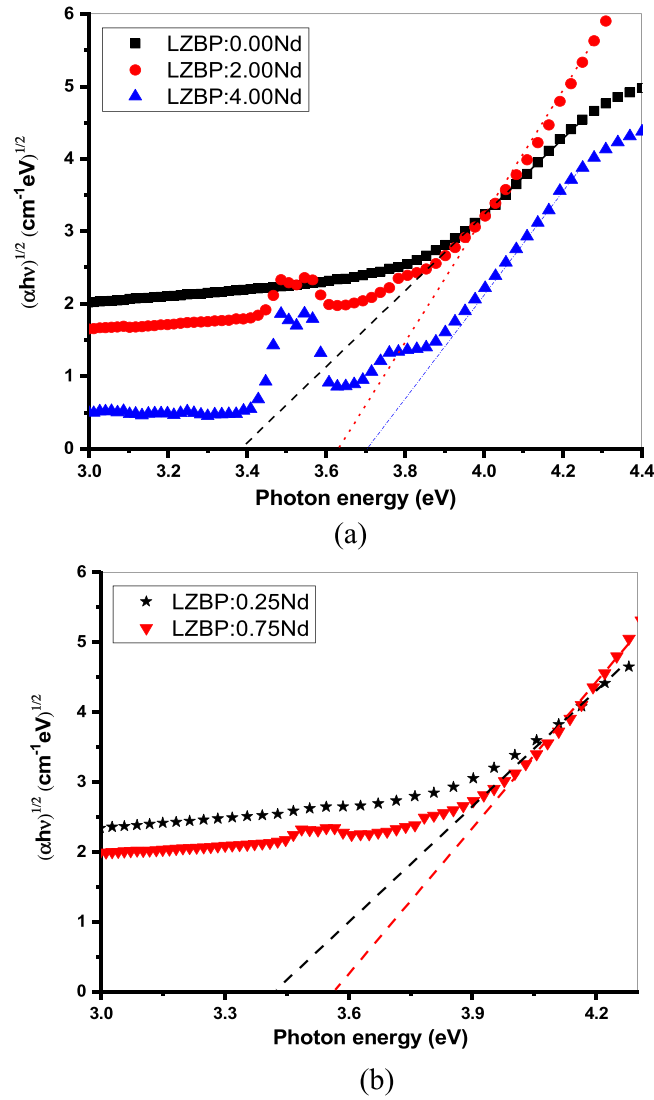


Fig. 6. (a). Variation of $(\alpha h\nu)^{0.5}$ with $h\nu$ of LZBP:xNd glasses, where $x = 0.0, 2$ and 4 wt%. (b). Variation of $(\alpha h\nu)^{0.5}$ with $h\nu$ of LZBP:xNd glasses, where $x = 0.25$ and 0.75 wt%.

$$A(\lambda) = \frac{B(hc)^{r-1}d}{2.303} \lambda \left(\lambda^{-1} - \lambda_g^{-1} \right)^r = K \lambda \left(\lambda^{-1} - \lambda_g^{-1} \right)^r \quad (13)$$

where $K = [B(hc)^{r-1}d/2.303]$ and its value is depending on the value of r .

Eq.(13) helps to compute the E_g using absorbance data and without the need of the d values of the samples. The λ_g can be estimated by the extrapolation of the linear region of $(A\lambda^{-1})^{r-1}$ against (λ^{-1}) at $(A\lambda^{-1})^{r-1} = 0$. Figs. 8 and 9 show the variation of $(A\lambda^{-1})^{1/2}$ and $(A\lambda^{-1})^2$ with (λ^{-1}) for I and D allowed transition, respectively. This method to estimated E_g is the absorption spectral fitting (ASF) and the E_g has been computed by:

$$E_{opt.}^{ASF,D,I} = 1239.83 \times \lambda_g^{-1} \quad (14)$$

Based on Fig. 10, when the Nd_2O_3 content increased from 0.0 to 4 wt%, the indirect E_g of glass samples were increased from 3.39 eV to 3.71 eV and from 3.36 eV to 3.66 eV according to the determined procedures from Tauc's method and ASF method, respectively. This increase is due to:

- i) It should be noted that the strength of the oxygen bond in the glass network affects the absorption edge, and the increase in the cross-link density within the glass matrix displaces the E_g to higher energies.

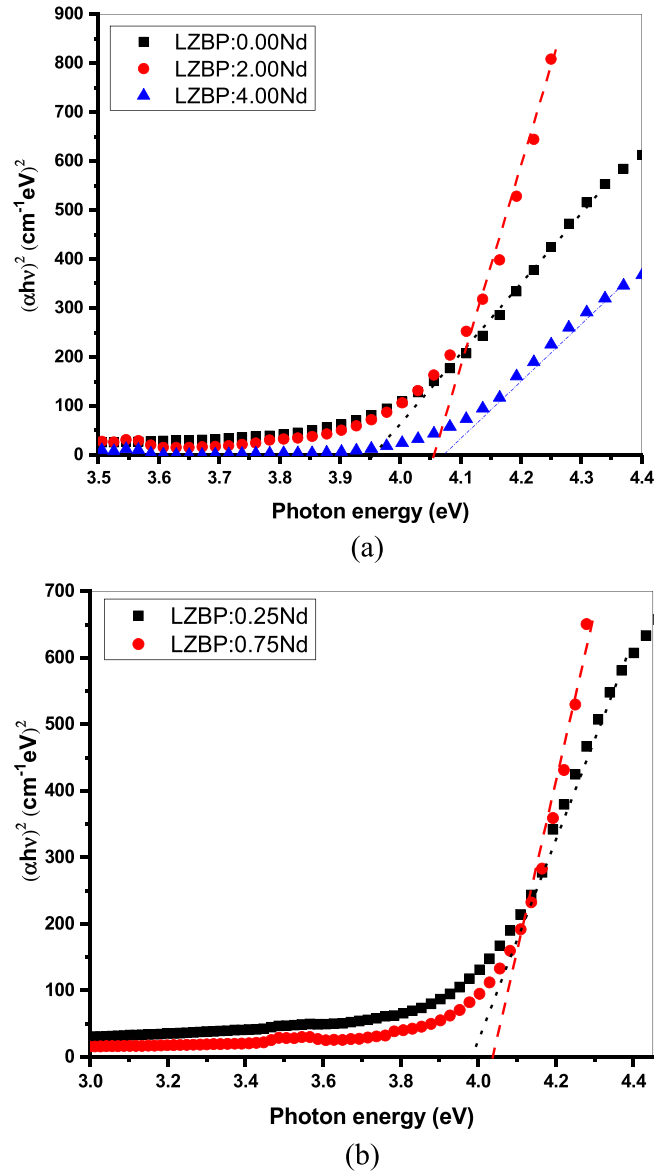


Fig. 7. (a). Variation of $(\alpha h\nu)^2$ with $h\nu$ of LZBP:xNd glasses, where $x = 0.0, 2$ and 4 wt%. (b). Variation of $(\alpha h\nu)^2$ with $h\nu$ of LZBP:xNd glasses, where $x = 0.25$ and 0.75 wt%.

- ii) The insertion of Nd^{3+} ions contributes to the polymerization process so that it strengthens the P-O-Nd linkage chain in the phosphate tetrahedron incorporated into the vitreous lattice. Also, the substitution of non-bridging oxygen ions with bridge oxygen reduces the ionic character of the oxygen species and the upper level of the valence band leading to an increase in the E_g .
- iii) The electronic shell of O^{2-} anions is affected by the addition of Nd^{3+} ions, and this leads to a modification of the polarizing of the cations.
- iv) The impurity band is extended due to the formation of the tail at the edges of the valence and conduction bands with the increase of Nd_2O_3 .
- v) The large ionic radius of Nd^{3+} ion which can reside in the distorted network through P-O-Nd Linkages.
- vi) The conversion of the BO_3 units to the BO_4 units which reduce the NBO at the expense of the BO and the rise of the anions number that are tightly bound to the host matrix. The rise of BO leads to the reduction in the donor center number as a result of the increase of E_g . The same behavior was reported in previous works [10].

Urbach's energy, E_u is attributed to the disorder in amorphous materials and can be estimated from Fig. 11 using the following relation:

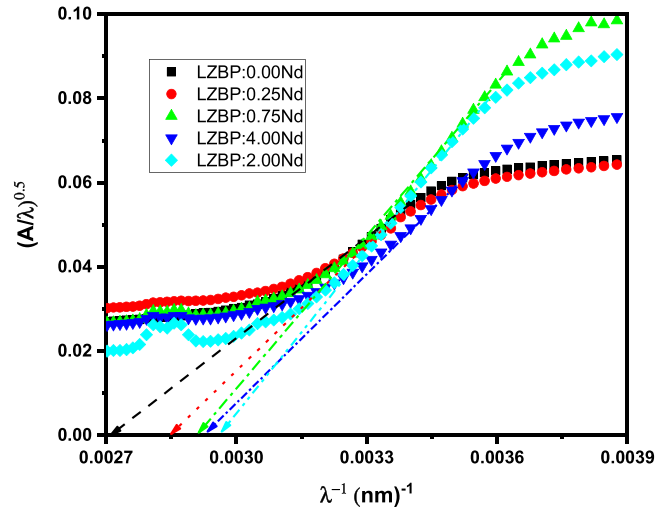


Fig. 8. Variation of $(A\lambda^{-1})^{0.5}$ with λ^{-1} for glass samples.

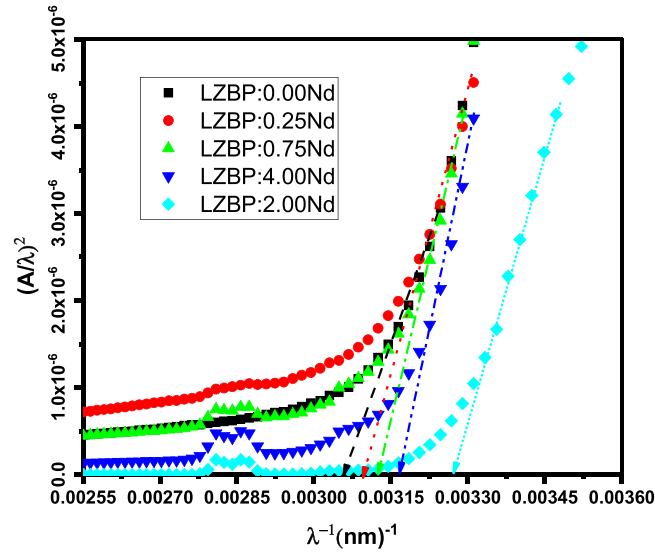


Fig. 9. Variation of $(A\lambda^{-1})^2$ with λ^{-1} for glass samples.

$$E_u = \frac{1239.83}{\text{slope}} \quad (15)$$

The E_u values are listed in Table 2. Urbach's energy lies in the range of the amorphous semiconductor. The decrease of E_u (0.170 – 0.088 eV) with the increase in Nd_2O_3 content may be due to the decrease of the localized states and the disorder degree in the band gap.

Steepness factor, S can be expressed as a function of E_u from:

$$S = \frac{K_B T}{E_u} \quad (16)$$

where T and K_B are the room temperature and Boltzmann's constant, respectively. The S values increase from 0.153 to 0.294 with the increase of the Nd_2O_3 content and the obtained S value has an opposite relation to the E_u as registered in Table 2.

The refractive index, n , can be evaluated from E_g values from [5]:

$$n = \left[\frac{3 - 2(E_g/20)^{1/2}}{(E_g/20)^{1/2}} \right]^{1/2} \quad (17)$$

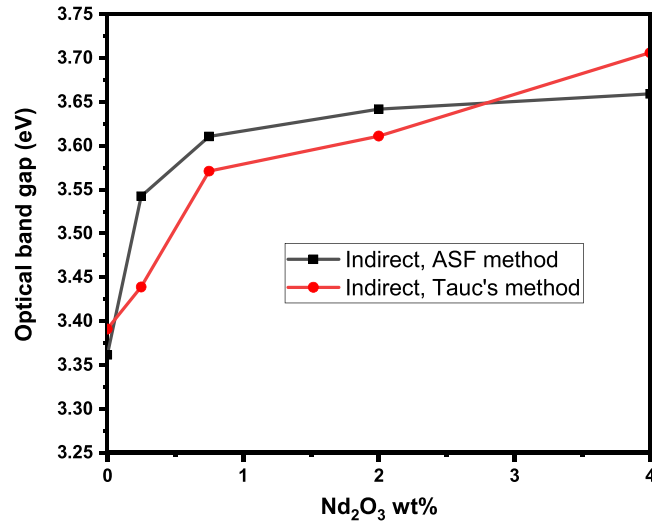


Fig. 10. Variation of $E_{opt.}^{ASF.I.}$ and $E_{opt.}^{Tauc's.I.}$ with Nd_2O_3 content.

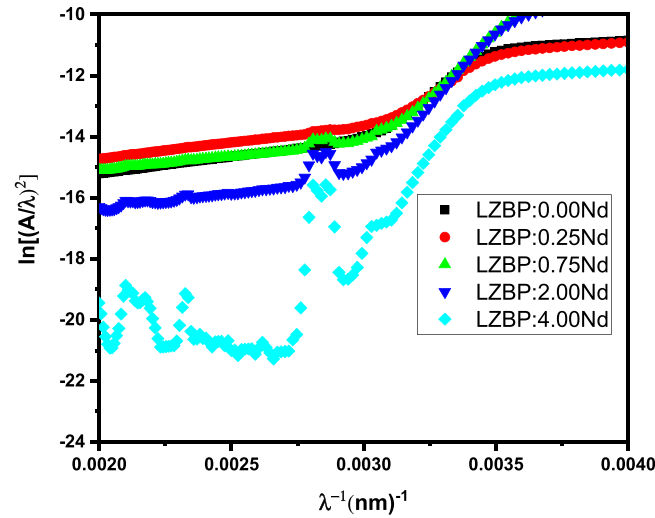


Fig. 11. Variation of $\ln(A/\lambda)^2$ with λ^{-1} for glass samples.

The values of n for all glass samples are listed in Table 2, which are in the range of 2.306–2.239 (by ASF method). The decreasing value of n with the increase of Nd_2O_3 content is the result of the variation in the density value which increases the light propagation inside the present glass because the increment of BO's and is due to the effect of Nd_2O_3 size and its concentration. The same trend was reported in [10].

The reflection loss (reflectivity), R_{loss} , from the glass surface and the optical transmission coefficient, $T_{opt.}$, of the LZBP:Nd glass samples are evaluated from the following relations [10]:

$$R_{loss} = \left[\frac{n-1}{n+1} \right]^2 \quad (18)$$

$$T_{opt.} = \frac{2n}{n^2 + 1} \quad (19)$$

The calculated values of R_{loss} and $T_{opt.}$ are given in Table 3. It is clear from the results that R_{loss} and $T_{opt.}$ have an inverse behavior with the Nd_2O_3 content, which agree with the theoretical expression where $T_{opt.}$ varies inversely proportional to the R_{loss} .

The optical dielectric constant, $\epsilon_{opt.}$, static dielectric constant, ϵ , and electrical susceptibility, $\chi_{elect.}$ were computed by the next relations [10]:

Table 2

The direct and indirect optical band gap by Tauc's method ($E_{opt.}^{Tauc's,D}$, $E_{opt.}^{Tauc's,I}$), The direct and indirect optical band gap by absorption spectrum fitting, ASF method ($E_{opt.}^{ASF,D}$, $E_{opt.}^{ASF,I}$), and refractive index, n by Tauc's and ASF methods, Urbach energy, E_u and steepness factor, S for glass samples.

Sample code	Optical band gap (eV) and Urbach Energy (eV)				Eu (eV)± 0.001	S ± 0.001	n by Tauc's method Indirect ± 0.001	n by ASF method Indirect ± 0.001
	Tauc's method, $E_{opt.}^{Tauc's}$		ASF method, $E_{opt.}^{ASF}$					
	Indirect, $E_{opt.}^{Tauc's.I} \pm 0.01$	Direct, $E_{opt.}^{Tauc's.D} \pm 0.01$	Indirect, $E_{opt.}^{ASF.I} \pm 0.01$	Direct, $E_{opt.}^{ASF.D} \pm 0.01$				
LZBP: 0.00Nd	3.39	3.96	3.36	3.79	0.170	0.153	2.299	2.306
LZBP: 0.25Nd	3.44	3.99	3.54	3.84	0.165	0.157	2.288	2.264
LZBP: 0.75Nd	3.57	4.03	3.61	3.87	0.144	0.180	2.258	2.250
LZBP: 2.00Nd	3.61	4.08	3.64	3.93	0.125	0.207	2.250	2.243
LZBP: 4.00Nd	3.71	4.09	3.66	4.06	0.088	0.294	2.229	2.239

Table 3

Reflection loss R_{loss} , optical transmission coefficient $T_{opt.}$, static dielectric constant ϵ , optical dielectric constant $\epsilon_{opt.}$, electrical susceptibility $\chi_{elect.}$, molar refraction R_m , molecular polarizability α_m and metallization criterion $M_{crit.}$ of prepared glass samples.

Sample code	$R_{loss} \pm 0.001$	$T_{opt.} \pm 0.001$	$\epsilon \pm 0.001$	$\epsilon_{opt.} \pm 0.001$	$\chi_{elect.} \pm 0.001$	$R_m, \text{cm}^3/\text{mol} \pm 0.001$	$\alpha_m, \text{\AA}^3 \pm 0.001$	$M_{crit.} \pm 0.001$
LZBP: 0.00Nd	0.156	0.730	5.318	4.318	0.344	23.106	9.169	0.409
LZBP: 0.25Nd	0.150	0.739	5.128	4.128	0.329	22.652	8.989	0.421
LZBP: 0.75Nd	0.148	0.742	5.061	4.061	0.323	22.425	8.899	0.425
LZBP: 2.00Nd	0.147	0.744	5.031	4.031	0.321	22.368	8.876	0.427
LZBP: 4.00Nd	0.146	0.745	5.014	4.014	0.320	22.508	8.932	0.428

$$\epsilon_{opt.} = \epsilon - 1 = n^2 - 1 \quad (20)$$

where ϵ is the static dielectric constant ($\epsilon = n^2$).

$$\chi_{elect.} = \frac{\epsilon - 1}{4\pi} \quad (21)$$

Values of $\epsilon_{opt.}$, ϵ and $\chi_{elect.}$ are listed in Table 3. From the results, the values of $\epsilon_{opt.}$ and ϵ are reduced from 5.318 to 5.014 and 4.318–4.014, respectively and the $\chi_{elect.}$ values are reduced with the increase of Nd_2O_3 content due to the decreasing number of NBOs.

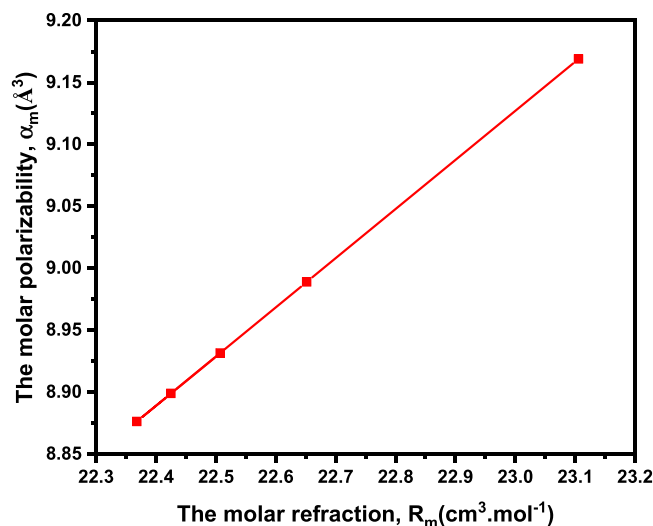


Fig. 12. Variation of α_m with R_m of glasses.

The molar refraction, R_m , and the molar polarizability, α_m , of the glass samples are calculated by [5,10]:

$$R_m = \frac{(n^2 - 1)}{(n^2 + 2)} V_m \quad (22)$$

$$\alpha_m = \frac{3}{4\pi N_A} R_m = R_m / 2.52 \quad (23)$$

Both R_m and α_m are found to be in the ranges of 23.106–22.368 cm³. Mol⁻¹ and 9.169–8.876 Å³ respectively, as listed in Table 3. The R_m relationship against the α_m is illustrated in Fig. 12 and the linear proportionality between them confirms the linear dependence of the R_m on the α_m .

Metallization criterion, $M_{crit.}$, of glasses is given by:

$$M_{crit.} = \frac{V_m - R_m}{V_m} \quad (24)$$

The metallic or non-metallic nature prediction of the materials depends on the following conditions: $R_m/V_m \geq 1$, for metallic nature, and $R_m/V_m < 1$, for insulating nature. The $M_{crit.}$ was found to increase from 0.409 to 0.428 with the rise in Nd₂O₃ content (Table 3). So that, the LZBP:Nd glasses have insulating nature.

3.4. Fourier transformed infrared (FTIR) spectra

The FTIR spectra have been recorded for all samples as shown in Fig. 13 and their assignments are listed in Table 4 in order to reveal the following defined bands:

- i. The bands within the region 477–487 cm⁻¹ are assigned to the bending vibration of O–P–O unit. These bands may be overlapped with both the ZnO tetrahedral bonds [11] and the stretching vibrations of Nd–O bond [12].
- ii. The band within the region 771–786 cm⁻¹ is assigned to symmetric stretching vibrations of P–O–P, ν_s (P–O–P) [13]. This band may be overlapped with another band ascribed to the bending vibrations of B–O–B bond in (BO₄) units [14].
- iii. The appearance of the band at nearly 916–931 cm⁻¹ is an indication of the presence of (P–O–P) as bridging oxygen bonds (BOs), ν_{as} (P–O–P) [15]. In addition, this band is due to the stretching vibrations of B–O bonds in [BO₄] tetrahedral units [16].
- iv. The band located at 1028–1066 cm⁻¹ is assigned to the asymmetric vibration of BO₄ [17].
- v. The bands in the region between 1076 and 1081 cm⁻¹ are related to the symmetrical stretching vibration of PO₂, ν_s (PO₂) [15], to asymmetric stretching vibrations of (PO₃)²⁻ and to P–O⁻ symmetric stretching [18].
- vi. The absorption band lying between 1264 and 1274 cm⁻¹ may be due to the stretching vibrations of P=O [19]. This band is superposed with both the asymmetric stretching vibrations of the two non-bridging oxygen atoms bonded to a phosphorus atom in a Q² phosphate tetrahedron, ν_{as} (PO₂) [20] and the symmetric stretching vibrations of B–O bonds in [BO₃] triangular units with NBO atoms [21].
- vii. The band at about 1331–1361 cm⁻¹ is related to the asymmetric stretching vibrations of (PO₂)⁻ in the (Q²) units [22] as well as the B–O stretching vibrations of trigonal [BO₃]³⁻ units [23].
- viii. The absorption bands observed around 1631 cm⁻¹ are due to the stretching and bending vibrations of P–O–H bridges and H₂O which is due to the absorption of moisture during the preparation of KBr pellets for infrared measurements [24].
- ix. The band at ~2358 cm⁻¹ is attributed to the antisymmetric stretching of the adsorbed water molecules [24].
- x. Doublet weak absorption bands observed in the ranges 2853–2858 cm⁻¹ and 2920–2935 cm⁻¹ are due to the presence of hydrogen bonds [24] and stretching modes of O–H bonds. The intensity of these bands decreases with the increase of the Nd₂O₃ content leading to the enhance of the lower non-radiative losses within current glass system.
- xi. The broad band observed in the range 3447–3495 cm⁻¹ is assigned to the O–H stretching vibrations of the hydroxyl group, molecular water. Also, this band may be due to B–O–H vibrations [24]. The intensities of these bands are found to decrease with the increase of the Nd₂O₃ content which indicate the decrease of the hygroscopic nature of the glasses (Fig. 13).

The deconvoluted spectra of the LZBP:4.00Nd glass sample is shown in Fig. 14 (as an example). Table 5 displays the band center, x_c , area, A , and full width at half maximum, w , of the component bands. The x_c , A and w of deconvoluted peaks change with the addition of Nd₂O₃. Based on deconvoluted data, the band center of the base glass (LZBP:0.00Nd) consists of eight peaks at about 848.43, 888.11, 932.78, 1002.7, 1083.7, 1266.6, 1338 and 1639.9 cm⁻¹. These bands are shifted to higher wavelengths with the addition of Nd₂O₃ especially at high concentration of Nd³⁺ ions. The bands at 848 cm⁻¹ may be due to B–O stretching vibrations of tetrahedral BO₄ units and its area increases with the addition of Nd₂O₃ and this showed that, the BO₃ units are converted to BO₄ units which led to the increase in the density of glasses because the BO₄ units are denser than the BO₃ units. Furthermore, the area of the band at ~1338 cm⁻¹, which is related to Q² units, decreases with increasing Nd₂O₃ content due to the conversion of these units (Q²) to other phosphate units such as Q¹ and/or Q⁰ units. Moreover, obvious changes can be observed in the band at 1083–1264 cm⁻¹ (vibrations of P–O⁻ groups) and its area is increased due to the increase of non-bridging oxygen within glass system with the addition of Nd₂O₃.

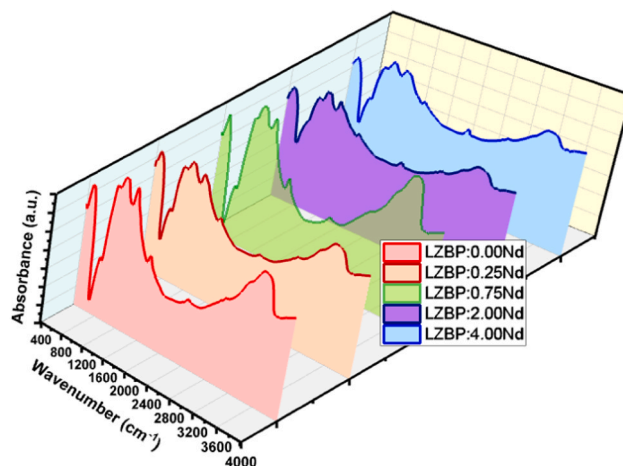


Fig. 13. FTIR absorption spectra of glass samples.

Table 4

Assignment of absorption bands in IR spectra of LZBP:Nd glasses.

Band position (cm ⁻¹)	Assignment	Refs.
477–487	ZnO tetrahedral bonds;	[11]
	Stretching vibrations of Nd-O bond.	[12]
771–786	Symmetric stretching vibrations of P-O-P, $\nu_s(\text{P-O-P})$;	[13]
	Bending vibrations of B-O-B bond in (BO ₄) units.	[14]
916–931	$\nu_{as}(\text{P-O-P})$	[15]
	Stretching vibrations of B–O bonds in [BO ₄] tetrahedral units.	[16]
1028–1066	Asymmetric vibration of BO ₄ .	[17]
1076–1081	Symmetrical stretching vibration of PO ₂ ²⁻ , $\nu_s(\text{PO}_2^{2-})$;	[15]
	Asymmetric stretching vibrations of (PO ₃) ₂ ²⁻ ;	[18]
	P-O ⁻ symmetric stretching.	[18]
1264–1274	Stretching vibrations of P ⁺ O ⁻ ;	[19]
	$\nu_{as}(\text{PO}_2)$;	[20]
	Vibrations of B-O bonds in [BO ₃] triangular units with NBO atoms.	[21]
1331–1361	Asymmetric stretching vibrations of (PO ₂) ⁻ in metaphosphate (Q ²) units;	[22]
	B–O stretching vibrations of trigonal [BO ₃] ³⁻ units.	[23]
1631	Stretching and bending vibrations of P–O–H bridges and H ₂ O.	[24]
~2358	Antisymmetric stretching of the adsorbed water molecules.	[24]
2853–2858 and 2920–2935	Hydrogen bonds;	[24]
	Stretching modes of O-H bonds.	
3447–3495	The O-H stretching vibrations of the hydroxyl group;	[24]
	Molecular water;	
	B–O–H vibrations.	

3.5. Gamma-ray shielding

To estimate the mass attenuation coefficient (MAC) of the samples under study for their potential use as gamma shielding, a Phy-X/PSD software program was used. The results of MAC is displayed in Fig. 15 as a function of photon energies from 0.015 to 15 MeV with respect to Nd₂O₃ content. From Fig. 15, it is noted that the increase in Nd₂O₃ content leads to the increase in MAC due to the increase in the mass density of the samples and due to the atomic numbers of the added element (Nd, Z = 60). The maximum values of MAC are 10.27, 10.45, 10.56, 10.76, 10.86, 11.63 and 12.57 cm²/g for LZBP:0.00Nd-LZBP:4.0Nd glasses at 15 keV photon energy, respectively. Following that the values of MAC begin to decrease until it reaches the lowest values at 15 MeV photon energy. This may be due to, at the lower energies of the photon, the photoelectric effect cross section is dominant at these energies and varies with $Z^4 \cdot E^{-3.5}$ [25]. Finally, from the results of the MAC, we conclude that an increase in the Nd₂O₃ content leads to an increase in the MAC as a result of the increase in the atomic number of the added element.

Fig. 16 shows the results of the half value layer (HVL) of the glass under study which depend on the energy of the photon and the Nd₂O₃ content. The HVL of the material depends on the linear attenuation coefficient as well as on the type of the material. It is also proportional to the energy of the photon and is inversely proportional to the density of the material. The lowest HVL of all samples are 0.027, 0.026, 0.026, 0.025, 0.025, 0.023 and 0.021 cm for LZBP:0.00Nd-LZBP:4.0Nd at 15 keV, respectively. From Fig. 16, we observe a gradual increase in the HVL with an increase in the energy of the photon, as well as a decrease in the HVL with an increase in the Nd₂O₃ content. The highest HVL was for LZBP:0.00Nd glass sample which means a less shielding capacity and the lowest HVL was

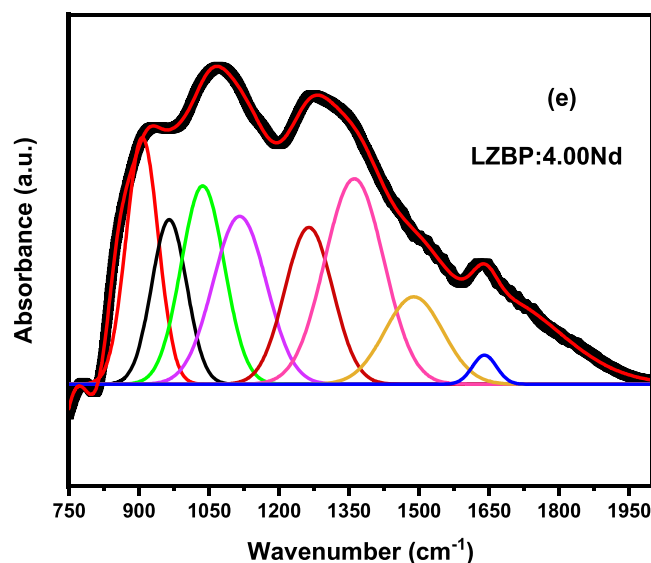


Fig. 14. Gaussian deconvoluted FTIR spectra for the LZBP:4.00Nd glass.

Table 5

Deconvolution parameter of the infrared spectra of samples, where x_c , A and w are the band center, an area and full width at half maximum of the component bands respectively.

Sample code		Peak position (x_c), area (A) and full width at half maximum (w)							
LZBP:0.00Nd	x_c	848.43	888.11	932.78	1002.7	1083.7	1266.6	1338	1639.9
	A	6.16	22.64	25.29	46.87	25	11.28	50.9	4.88
	w	36.90	62.91	73	105.5	97.3	67.79	153.2	63.39
LZBP:0.25Nd	x_c	844.68	887.24	963.92	1052.2	1133.5	1253.6	1351.2	1641.9
	A	7.56	56.57	69.85	70.17	50	71	120	3.44
	w	43.15	95.55	116.7	111.4	111.4	393.5	270.9	56.1
LZBP:0.75Nd	x_c	848.34	911	927	1033.2	1096.8	1264.9	1314.7	1636.9
	A	0.32	4.41	88.4	52.47	18.6	12.77	108	6.78
	w	31.94	57	168	111	80.3	74	224	62.54
LZBP:2.00Nd	x_c	902.79	926.75	969	1021.9	1086.4	1260.7	1316.8	1640.7
	A	29	0.43	3.5	11	19.5	8.97	10.6	2
	w	143	37	56.9	68.6	91.6	70	96.5	81.3
LZBP:4.00Nd	x_c	907	964.8	1036.7	1115.8	1264	1361.4	1489	1640.3
	A	40	29.5	43.8	45.9	38.44	62	26.2	3.6
	w	80.5	88	108.6	134.6	120.6	148.6	147.2	61

for LZBP:4.00Nd glass sample which means a high shielding capacity.

The effective atomic number (Z_{eff}) is a useful parameter especially in biology and medical applications. The concept of Z_{eff} is based on the basic theory of X-ray and γ -ray interactions with matter. The concept of Z_{eff} is used in composite materials of more than one element and is varied with energy. The results of Z_{eff} for the glass samples under study as a function of photon energy with respect to Nd_2O_3 content are shown in Fig. 17. The maximum Z_{eff} values are found to belong to the LZBP:4.00Nd glass while the lowest values are for LZBP:0.00Nd glass as a result of the highest and lowest molecular weight of the samples. The maximum values of Z_{eff} for LZBP:0.00Nd-0.75LZBP:0.75Nd glasses are 18.04, 18.29 and 18.73 at 0.015 MeV, and 20.40 and 20.06 at 0.05 MeV for LZBP:2.00Nd and 4.00LZBP:0.75Nd glasses, respectively. The lowest values are 9.14, 9.16, 9.19, 9.29 and 9.39 which are found at 1 MeV for LZBP:0.00Nd-LZBP:4.00Nd glasses, respectively. At energies higher than 1 MeV, the effective atomic number increases again as the dominant pair production interaction.

Fig. 18 shows the electron density N_{eff} for all the glass samples as a function of the incident photon energy. LZBP:4.00Nd sample shows the maximum value for N_{eff} whereas the minimum value for N_{eff} was observed for LZBP:0.00Nd in the 0.015–15 MeV energy region. The variation in N_{eff} with the incident photon energy has the same shape as what was observed for the Z_{eff} number. It is found that N_{eff} for the prepared glasses has its maximum value at 0.015 MeV and is equal to $(5.81, 5.88, 6.0, 6.32, \text{ and } 6.64) \times 10^{23}$ electron/g for LZBP:0.00Nd- LZBP:4.00Nd glasses, respectively. Also, it is observed from Fig. 18 that the N_{eff} initially decreases as the energy of the photon increases up to 1.0 MeV except at 0.05 MeV due to the k-edge of Nd doping the glasses. It was found to increase due to the pair production interaction in the high photon energy range.

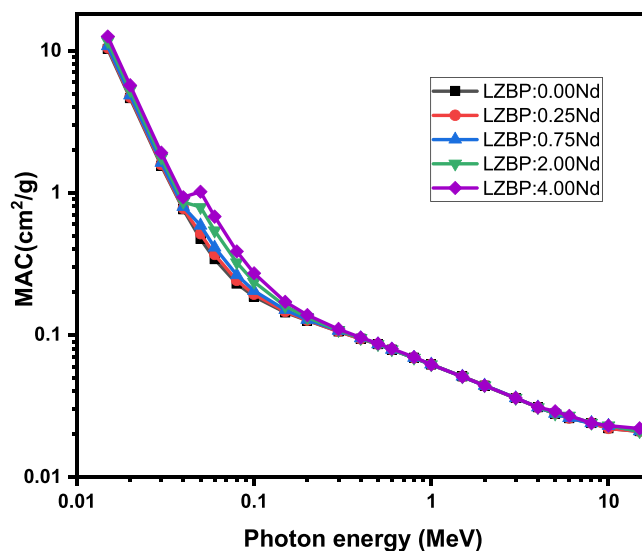


Fig. 15. Mass attenuation coefficient values of LZBP:0.00Nd-LZBP:4.00Nd glasses at different photon energies.

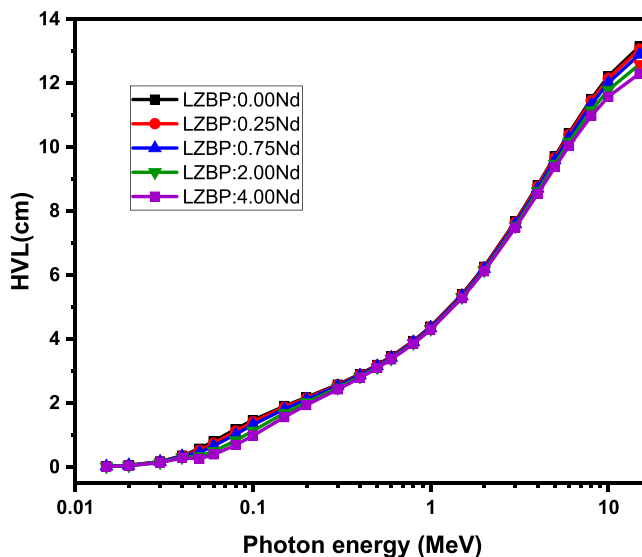


Fig. 16. Half value layer values of LZBP:0.00Nd-LZBP:4.00Nd glasses at different photon energies.

4. Conclusions

The Nd^{3+} ions effects on the investigated glasses are summarized in the following points:

1. The density of the samples was increased from 2.528 to 2.588 g.cm^{-3} , while the molar volume was decreased to 38.99 $\text{cm}^3.\text{mol}^{-1}$ and beyond 0.75 wt% was increased with the increment Nd_2O_3 content.
2. The T_g of the samples was increased from 378.11 to 420.15 $^\circ\text{C}$ after the addition of Nd_2O_3 .
3. The presence of several structural units such as PO_4 , BO_3 and BO_4 appeared within the glass matrix.
4. Based on Tauc's method, the $(E_{\text{opt}}^{\text{ASF,I}})$ values were in the range of (3.36–3.66 eV), while the $(E_{\text{opt}}^{\text{ASF,D}})$ values were in the range of (3.79–4.06 eV).
5. Based on ASF method, the $(E_{\text{opt}}^{\text{ASF,I}})$ values were in the range of (3.39–3.71 eV), while the $(E_{\text{opt}}^{\text{ASF,D}})$ values were in the range of (3.96–4.09 eV).
6. linear relation between α_m and R_m was obtained and the values of ϵ and ϵ_{opt} decreased with the increase in Nd_2O_3 content.
7. The $M_{\text{crit.}}$ values were in the range of 0.409–0.428, which indicate that the investigated glass expose non-metallic nature.

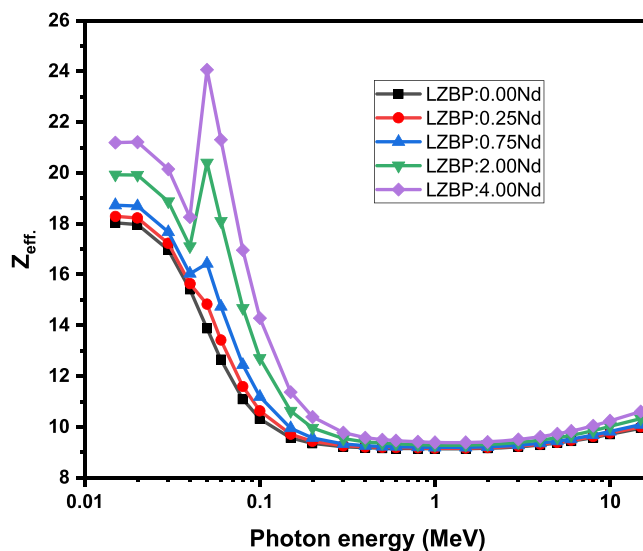


Fig. 17. Effective atomic numbers of LZBP:0.00Nd-LZBP:4.00Nd glasses at different photon energies.

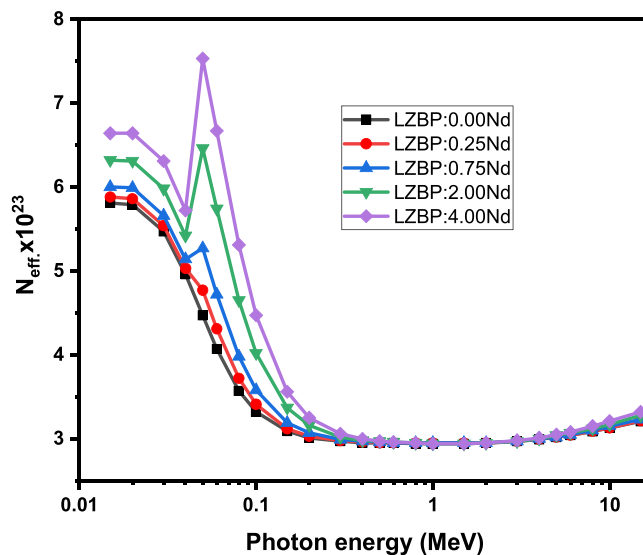


Fig. 18. Effective electron density of LZBP:0.00Nd-LZBP:4.00Nd glasses at different photon energies.

8. The glass LZBP:4.0Nd has the lowest HVL and thus has a high shielding capacity for gamma rays.

Declaration of Competing Interest

The author declares no conflict of interest.

Acknowledgements

M. A. Algradee is very much thankful to Cairo University, Egypt for providing laboratory facilities and to IIE, New York for granting SRF fellowship.

References

- [1] R. El-Mallawany, *Tellurite Glasses Handbook: Physical Properties and Data*, CRC Press, Taylor & Francis Group, Boca Raton, 2012.
- [2] V.A. Adhwaryu, D.K. Kanchan, Ag⁺ ion conduction in AgI-Ag₂O-B₂O₃-P₂O₅ glass electrolyte, *Mater. Sci. Eng. B* 263 (2021), 114857.

- [3] Kyeong Dae Park, Karam Han, Yong Gyu Choi, Woon Jin Chung, Chemical strengthening of sodium aluminosilicate glasses containing P_2O_5 and B_2O_3 , *J. Non Cryst. Solids* 554 (2021), 120600.
- [4] M.I. Sayyed, R. El-Mallawany, Shielding properties of $(100-x)TeO_2-(x)MoO_3$ glasses, *Mater. Chem. Phys.* 201 (2017) 50–56.
- [5] M. Algradee, M. Sultan, O.M. Samir, A. Elwhab, B. Alwany, Electronic polarizability, optical basicity and interaction parameter for Nd_2O_3 doped lithium–zinc–phosphate glasses, *Appl. Phys. A* 123 (524) (2017) 1–12.
- [6] M.A. Algradee, A. Elwhab, B. Alwany, M. Sultan, M. Elgoshimy, Q. Almoraisy, Physical and optical properties for Nd_2O_3 doped lithium–zinc–phosphate glasses, *Optik* 142 (2017) 13–22.
- [7] E. Sakar, Ö.F. Özpolat, B. Alım, M.I. Sayyed, M. Kurudirek, Phy-X / PSD: Development of a user friendly online software for calculation of parameters relevant to radiation shielding and dosimetry, *Radiat. Phys. Chem.* 166 (2020), 108496.
- [8] N.F. Mott, E.A. Davis, *Electronic Process in the Non Crystalline Materials*, second ed., Clarendon Press/Oxford University, New York, 1979.
- [9] L.E. Alarcon, A. Arrieta, E. Camps, S. Muhl, S. Rudil, E.V. Santiago, An alternative procedure for the determination of the optical band gap and thickness of amorphous carbon nitride thin films, *Appl. Surf. Sci.* 254 (2007) 412–415.
- [10] Sangeeta B. Kolavekar, N.H. Ayachit, Impact of Pr_2O_3 on the physical and optical properties of multi-component borate glasses, *Mater. Chem. Phys.* 257 (2021), 123796.
- [11] R.K. Brow, Review: the structure of simple phosphate glasses, *J. Non Cryst. Solids* 263–264 (2000) 1–28.
- [12] L.M. Sharaf El-Deen, M.S. Al Salhi, M.-M. Elkholy, IR and UV spectral studies for rare earths-doped tellurite glasses, *J. Alloy. Compd.* 465 (2008) 333–339.
- [13] M. Abid, M. Et-Sabiron, M. Taibi, Structure and DC conductivity of lead sodium ultraphosphate glasses, *Mater. Sci. Eng. B* 97 (1) (2003) 20–24.
- [14] H.A. ElBatal, A.M. Abdelghany, I.S. Ali, Optical and FTIR studies of CuO -doped lead borate glasses and effect of gamma irradiation, *J. Non Cryst. Solids* 358 (2012) 820–825.
- [15] L. Koudelka, P. Mošner, Borophosphate glasses of the $ZnO-B_2O_3-P_2O_5$ system, *Mater. Lett.* 42 (2000) 194–199.
- [16] E.I. Kamitsos, A. Patsis, G.D. Chryssikos, Infrared reflectance investigation of alkali diborate glasses, *J. Non Cryst. Solids* 152 (1993) 246–257.
- [17] C. Gautam, A.K. Yadav, A.K. Singh, Associated factors for uterine prolapse, *J. Nepal Health Res. Counc.* 10 (2012) 1–4.
- [18] Raja Junaid Amjad, Tasso O. Sales, Abdul Sattar, Carlos Jacinto, M. Reza Dousti, Spectral studies of highly Dy^{3+} doped $PbO-ZnO-B_2O_3-P_2O_5$ glasses, *J. Lumin.* 231 (2021), 117839.
- [19] S. Ibrahim, M.A. Baki, F. El-Diasty, Zinc borophosphate glasses for infrared based optical applications, *Opt. Eng.* 51 (2012), 093401,1–7.
- [20] S. Kabi, A. Ghosh, Mixed glass former effect in AgI doped silver borophosphate glasses, *Solid State Ion.* 262 (2014) 778–781.
- [21] H.D. Subhashini, Shashikala, N.K. Udayashankar, Influence of Fe^{3+} ions on optical, structural, thermal and mechanical properties of $Li_2O-Na_2O-K_2O-ZnO-B_2O_3$ based glass system, *Ceram. Int.* 46 (2020) 5213–5222.
- [22] F. Wang, Q.L. Liao, K.R. Chen, S.Q. Pan, M.W. Lu, Glass formation and FTIR spectra of CeO_2 -doped $36Fe_2O_3-10B_2O_3-54P_2O_5$ glasses, *J. Non Cryst. Solids* 409 (2015) 76–82.
- [23] Manju Bala, Sunil Agrohiya, Sajjan Dahiya, Anil Ohlan, R. Punia, A.S. Maan, Effect of replacement of Bi_2O_3 by Li_2O on structural, thermal, optical and other physical properties of zinc borate glasses, *J. Mol. Struct.* 1219 (2020), 128589.
- [24] Y.M. Lai, X.F. Liang, S.Y. Yang, J.X. Wang, L.H. Cao, B. Dai, Raman and FTIR spectra of iron phosphate glasses containing cerium, *J. Mol. Struct.* 992 (1–3) (2011) 84–88.
- [25] A. Alalawi, M.S. Al-Buriahi, Y.S. Rammah, Radiation shielding properties of PNCKM bioactive glasses at nuclear medicine energies, *Ceram. Int.* 46 (2020) 15027–15033.